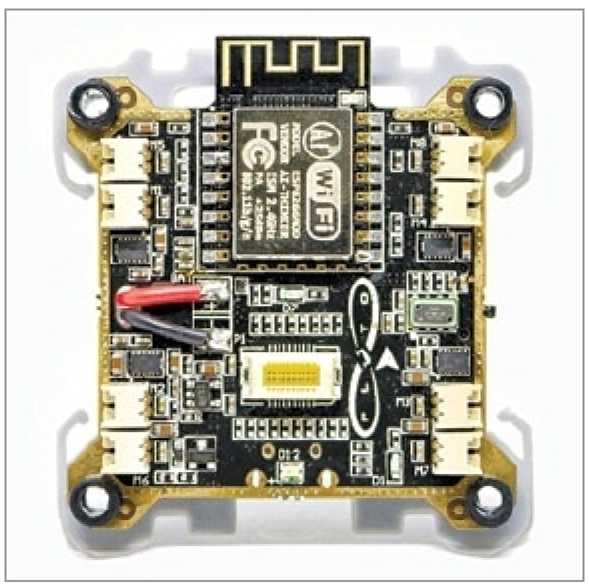
The Indiegogo campaign

Looking at interesting customisations and innovative use cases the tinkerers have built with their prototypes, Drona Aviation went ahead with Indiegogo campaign to reach out to people at a global scale. Leveraging the global reach of the platform, the team was able to connect with influencers, laboratories, professors and technical hobbyists across the world, receiving feedback and improvement tips on their kit and creating a reputation. Godbole says, "Initially, when we were selling Pluto, there was a significant lack of market communication. Through the campaign, we were able to find an interesting set of customers who were buying our products. It also helped us plan our future roadmap."

Receiving full crowdfunding through the campaign, PlutoX is available for pre-order on Indiegogo website, and will be reaching customers starting December 2018.

PlutoX parts



PrimusX board

be accessed later. The drone can also be programmed to perform image processing. Features like object following and colour recognition, among others, can be added as well. A lithium-polymer battery is used, which allows a flight time of up to 15 minutes.

PrimusX has an ESP microchip, which creates its own Wi-Fi hotspot. Wi-Fi-enabled devices like smartphones or controllers can connect to the drone using this network.

The mobile app has to be installed in the control device, which is compatible with both Android and iOS. It can also act as a troubleshooting device. Users can access sensor data and video feed using the app, configure the drone and modify its settings.

The drone can be programmed using an API-based software development kit (SDK) on the team's proprietary Cygnus IDE, which shortens the time taken in creating operational algorithms. Godbole says, "Custom-programming a drone for a specific task may require anywhere between 100 and 1000 lines of code. APIs in PlutoX bring that down to just 20 lines, delivering accurate performance."

The program is written in C++, and can be used to create any scale of operation. PlutoX uses Eclipse IDE to create the SDK.

Drona Aviation is manufacturing and assembling the kits in India, working with manufacturers in Mumbai and Gandhinagar. And it is importing batteries, motors and microcontrollers.

Production obstacles

Sourcing was a challenge in the Indian ecosystem. Godbole says, "While we knew how to make the technology work, finding each component at the right price was a major challenge. For example, to find the right camera, we initially had to go to China, in addition to a lot of market research in India."

Even the dearth of a proper manufacturing ecosystem caused hardships. Godbole recalls, "We reached out to many PCB makers. But the quality that most of them were delivering was disappointing. We lost about three months finding the right partner."

During the Indiegogo campaign, building strong market communication around the product was challenging. However, Godbole and team agrees that it was a learning experience, which further strengthened their marketing and technical skills.

—Paromik Chakraborty, technical journalist, EFY

AI-Based Solution For Personalised Styling

To help retail stores deliver personalised styling suggestions to customers, Sandeep Chatterjee, founder and chief executive officer at FR Tech Innovations, came up with the idea of iLUK. It is a computer vision and artificial intelligence (AI)-based solution. Chief technical officer Ravi Kiran Katha is responsible for technology implementation.

This is how the technology works. iLUK pod is placed in a retail outlet. The customer enters the pod, and gets photographed and scanned from different angles, using a set of RGB cameras. Once the cameras capture the images of the customer, these are passed through custom algorithms to construct a 3D model, or *avatar*, of the customer. It encrypts the 3D model

iLUK pod